

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	BS-STAT1
Program Name	Bachelor of Science in Statistics
Assessment Year	2025
Program Assessment Coordinator	Despina Stasi, Associate Teaching Professor, Applied Mathematics Department
Year of Current Assessment Plan	Y2
No Assessment This Year per Plan	☐
LO(s) Assessed	LO #3 SWBA to propose statistical models, evaluate their validity, use them to draw conclusions, and demonstrate understanding of their limitations.
Presented at program authority meeting date: <i>Please attach minutes</i>	October 15, 2025

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations **M**, **P** Proficient}

Learning Goal Assessed: LO #3 SWBA to propose statistical models, evaluate their validity, use them to draw conclusions, and demonstrate understanding of their limitations.

Semester in which artifacts were collected: Spring 2025

Artifact Source Course and Instructor: Math 446/546, Stasi

Sample size: Two(2) program students

Name of Rubric: Stasi2025

Artifact: Q4 parts a,b,c(i),c(ii) from the course final exam.

Concordance with LO#3:

In parts a and b of the question the students are asked to examine a time series and its ACF and PACF plots, discuss notable properties, and decide whether a particular statistical model is appropriate for this dataset and explain why. In part c the students are asked to evaluate model fit based on residual plots and explain how to use such tools in general and specifically for this problem. Thus the question requires

the students to propose appropriate statistical models, evaluate the validity of said models, and explain why others are not appropriate, use the model to draw conclusions and demonstrate understanding of the limitations of statistical models.

Rubrik:

Point Score	Assessment Category	Assessment	Explanation
0-4	D	<i>Does not meet expectations</i>	<i>Student does not demonstrate most of the following:</i> <i>- correctly identifies all key properties of the time series data in part a)</i> <i>- correctly concludes that ARMA model would not be suitable and that an appropriate transformation would be required first in part b)</i> <i>- correctly identifies the properties of white noise sequence expected in general in residual plots in part c(i) and</i> <i>- correctly evaluates the particular plots accordingly coming to the conclusion that the model is appropriate here in part c(ii).</i>
5-6	M	<i>Meets expectations</i>	<i>Student demonstrates most of the following:</i> <i>- correctly identifies all key properties of the time series data in part a)</i> <i>- correctly concludes that ARMA model would not be suitable and that an appropriate transformation would be required first in part b)</i> <i>- correctly identifies the properties of white noise sequence expected in general in residual plots in part c(i) and</i> <i>- correctly evaluates the particular plots accordingly coming to the conclusion that the model is appropriate here in part c(ii).</i>
7-8	P	<i>Proficient</i>	<i>Student demonstrates all or almost all the following:</i> <i>- correctly identifies all key properties of the time series data in part a)</i> <i>- correctly concludes that ARMA model would not be suitable and that an appropriate transformation would be required first in part b)</i> <i>- correctly identifies the properties of white noise sequence expected in general in residual plots in part c(i) and</i> <i>- correctly evaluates the particular plots accordingly coming to the conclusion that the model is appropriate here in part c(ii).</i>

Results

Artifact samples

Attach artifact samples. You can paste them in here if feasible, attach them to the end of this report, or submit them as supplemental files when submitting your reports. For data sets with ≤ 10 students, provide 1 in each category, D,M,P, (unless the category is empty, if so state that). For programs with ≤ 20 , provide 2. For other programs provide 3. Please redact any personally identifiable information, PII (which includes names and A#s). Please convert your documents to PDFs.

Please use the following naming convention for your artifact files: **AY25_CIM-CODE_Name of Artifact.pdf**.

Artifacts attached.

AY25_BS-STAT1_Artifact_Student1.pdf

AY25_BS-STAT1_Artifact_Student2.pdf

Data table (LO#3)

For each LO and provide a table pasted in from the downloadable [provided template](#), placing each student into the assessment categories (minimally {D, M, P}). You should collect and analyze data linked to student A#s locally, but please redact any personally identifiable information, PII (including names or A#s) in tables presented in this report. Provide a summary graph from the template presenting this distribution. You should also present additional analysis as deemed appropriate by the results seen, and the needs of the program.

Assessment Student ID	Point Score	Assessment Category	Score Explanation
1	7.5	P	The student correctly noted that the first model would not be appropriate without a transformation and that the residuals indicated a good model fit. The student noted most of the important properties of the time series, acf and pacf plots in part a, as well as the residual plots in part c but omitted to note seasonality being a factor in part a.
2	7	P	The student correctly noted that the first model would not be appropriate without a transformation and that the residuals indicated a good model fit. The student noted most of the important properties of the time series, acf and pacf plots in part a, as well as the residual plots in part c but omitted to note trend being a factor in part a and verifying constant variance in part c.

Description of findings

Summarize the discussion that occurred within the program authority when this was presented. Provide a short narrative interpreting the findings. Are there any areas of concern?

The number of program students taking the course in the Spring semester was small, but they were both Proficient in this assessment, which gives us 100% proficiency.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

At this particular iteration we achieved 100% Proficiency so no particular CI plans are indicated out of these results. However the department faculty consistently consider improvements to our programs and courses, including the BS, Statistics program.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

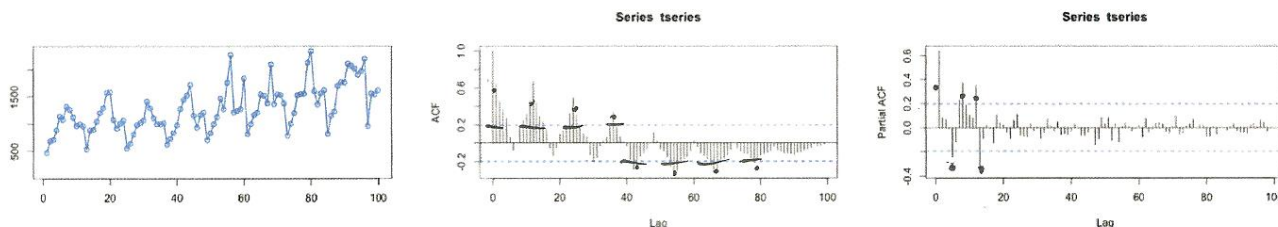
Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☒ After consideration and presentation to the program authority, no CI plans are recommended.

The current results do not indicate any particular CI plans. However the department faculty consistently consider improvements to our programs and courses, including the BS, Statistics program.

(12 pts) 4. Consider the following time series.



- (a) (2 pts) Comment on the figures above as they relate to time series analysis. What can you tell about the series from these? (for example could the series be white noise? are there other noteworthy patterns of behaviour? 2-3 sentences suffice)

There seems to be some autocorrelation in this series, the ACF and PACF plots are suggesting this. The series does not seem to be WN, with a slightly upward trend as time goes on.

15/2

Seasonality

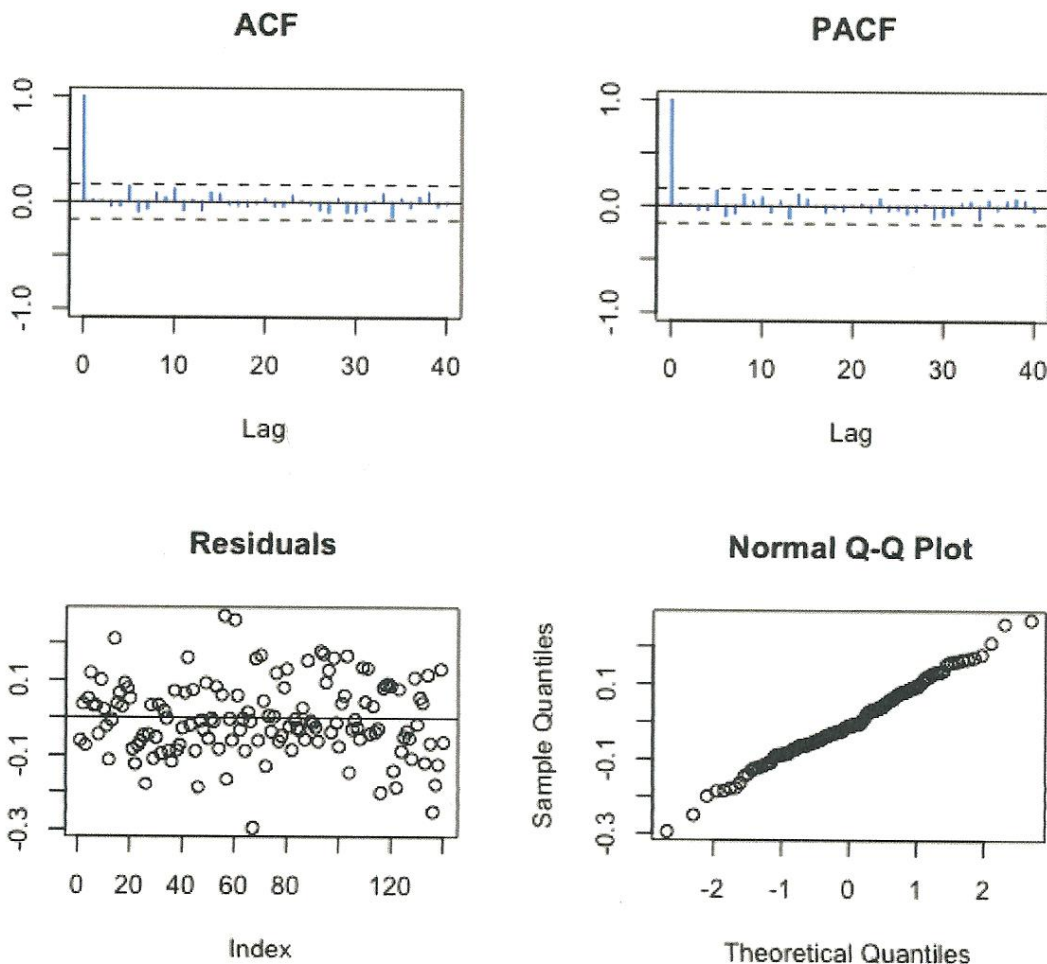
- (b) (2 pts) Is it appropriate to fit an ARMA model directly to this data? If yes, state why. If no, how might you try to alter your time series data to make ARMA modeling more likely to succeed.

It would not be suitable to fit an ARMA without the proper transformation. This time series would need to be verifiable WN before fitting an ARMA.

2

3.5

- (c) Suppose you have appropriately fit an ARMA model to some time series data and obtained the following residuals plots.



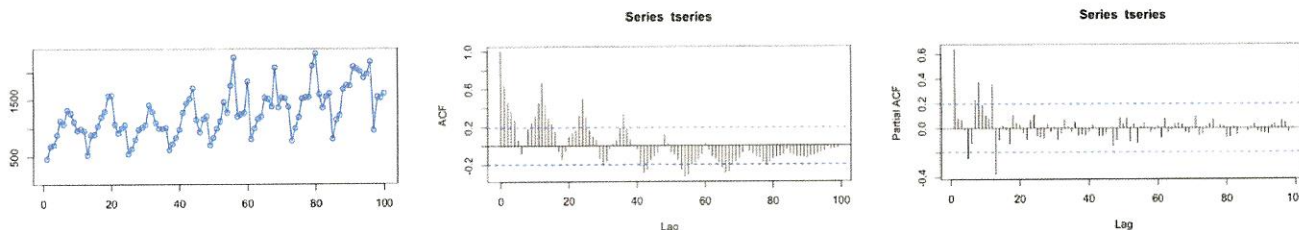
- (i) (2pts) Explain how you use each of the four residual plots to evaluate your ARMA model fit.

ACF - To check for autocorrelation in each lag ✓
 PACF - check for autocorrelation
 Residuals - check if residuals are centered around zero and approx. normal
 Q-Q - Check for residuals of the variance to steadily increase without any spikes or heavy tails. ✓

- (ii) (2pts) What do the plots indicate for this particular model fit?

Residuals are approximately normal around zero with no signs of autocorrelation ✓

(12 pts) 4. Consider the following time series.



- (a) (2 pts) Comment on the figures above as they relate to time series analysis. What can you tell about the series from these? (for example could the series be white noise? are there other noteworthy patterns of behaviour? 2-3 sentences suffice)

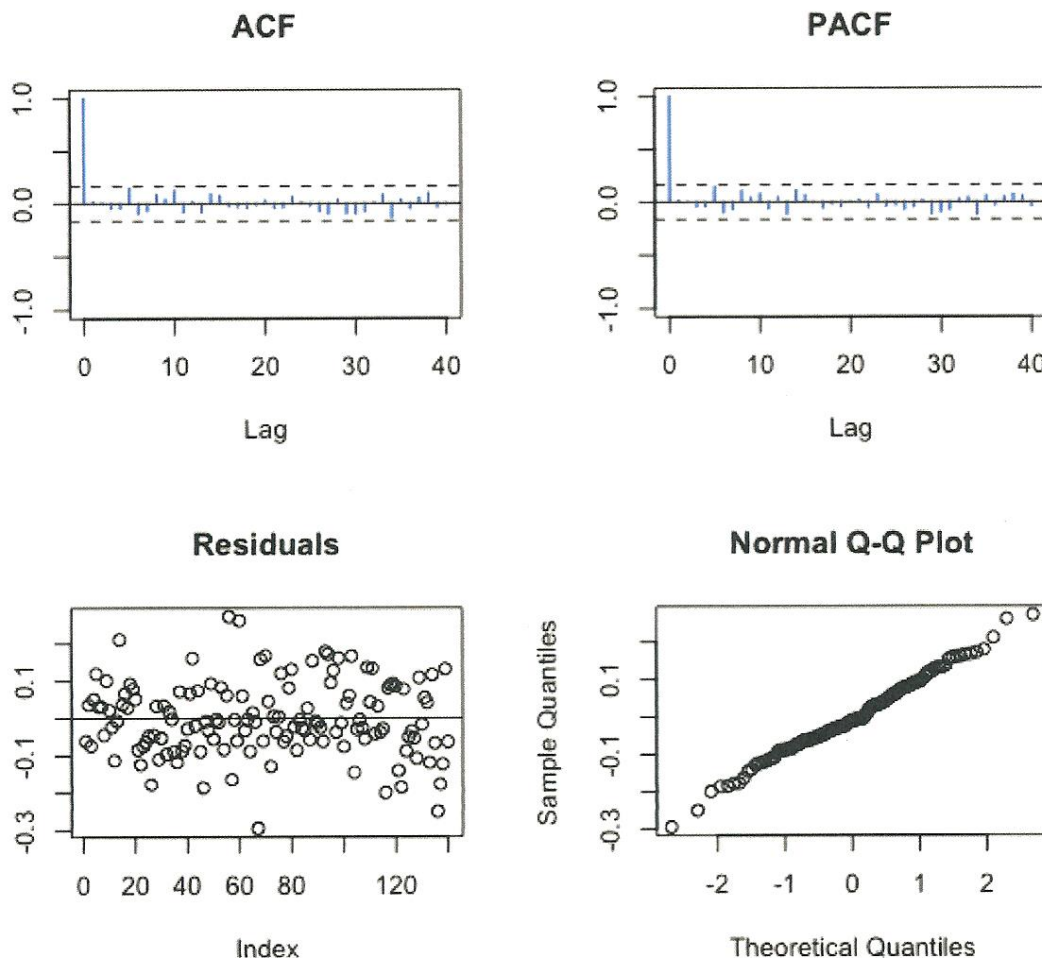
There is ~~seasonality~~ in the data for every ≈ 12 units in the first plot. Some lags in the ACF plot are outside the confidence bound, this indicates that there may be correlation issue within the data.

+ trend

- (b) (2 pts) Is it appropriate to fit an ARMA model directly to this data? If yes, state why. If no, how might you try to alter your time series data to make ARMA modeling more likely to succeed.

No, we would need to apply differencing (~~1st~~ order) to remove seasonality in the data before fitting an ARMA model.

- (c) Suppose you have appropriately fit an ARMA model to some time series data and obtained the following residuals plots.



- (i) (2pts) Explain how you use each of the four residual plots to evaluate your ARMA model fit.

Both ACF and PACF plots have lags within the confidence bands, therefore there is no autocorrelation problems for this ARMA model. The points in the residual plot are fairly balanced around zero, this indicates that the residuals are normalized. The points in the Q-Q plot form a fairly linear/straight line. These two plots indicate that the residuals behave like white noise.

- (ii) (2pts) What do the plots indicate for this particular model fit?

The plots indicate that the ARMA model is valid and reliable. The ARMA model is well-fitted.